ARNAV VASHISHTHA

Agentic AI Engineer · Full-Stack Engineer, AI/ML · Data & ML Engineering

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SUMMARY

Software engineer and data scientist who owns problems end to end — raw data collection, pipeline engineering, model development, and deployed user-facing applications. At NXP Semiconductors, shipped an LLM-powered normalization pipeline onboarding 2M+ previously unreportable records at ~95% automated resolution with ~60% lower LLM token cost, and architected a lead-time intelligence engine consolidating 60+ vendor files across 8 global commodities into automated Power BI reporting with predictive supplier-risk signals. Deep agentic experience — multi-agent orchestration, text-to-SQL agents, tool calling, RAG, human-in-the-loop applications.

TECHNICAL SKILLS

Languages & Core: Python, SQL, JavaScript/TypeScript, Java, SAP ABAP, PySpark, REST APIs, Postman, Git, Linux, DSA

ML & Deep Learning: PyTorch, Hugging Face Transformers, LLM fine-tuning (LoRA/QLoRA), distillation, scikit-learn, XGBoost, LightGBM, NLP, UMAP, HDBSCAN, model evaluation

Agentic & LLM Systems: LangChain, LangGraph, CrewAI, Google Gemini, multi-agent orchestration, tool use / function calling, text-to-SQL agents, RAG pipelines, prompt engineering, Pydantic structured output, vector databases, human-in-the-loop

Data & Analytics: pandas, NumPy, SciPy, Databricks, ETL/ELT, star-schema modelling, data warehousing, SQLite/MySQL, Power BI, Plotly, Advanced Excel, statistical analysis, fuzzy matching

Automation & Enterprise: Power Automate, Power Apps, Copilot Studio, Databricks Genie, SharePoint, Azure, SAP (FBL1N, BSIK/BSAK, LFA1, OData), Procure-to-Pay, workflow automation

Building & Shipping: Streamlit, FastAPI/Flask, Docker, CI / GitHub Actions

PROFESSIONAL EXPERIENCE

AI/ML Intern — NXP Semiconductors

Jan 2026 – Jul 2026 · Both workstreams scoped, built and deployed independently, end to end

P-Card Data Normalization — LLM-Powered Data Pipeline

- Built an LLM-driven cleaning and vendor-normalization pipeline that onboarded **2M+** historical Purchase-Card transactions into enterprise reporting — data previously excluded from analytics due to ambiguous, duplicated vendor records in raw bank feeds.
- Profiled in Excel how vendor names were entered across offices before designing resolution logic; resolved noisy strings via LLM reasoning with an integrated web-search API, then fuzzy-matched against the SAP vendor master — **~95%** automated resolution, with a Streamlit human-in-the-loop review app for the remaining ambiguous cases.
- Engineered a historical-resolution cache reusing confirmed mappings to cut LLM token cost **~60%**; deployed on Databricks (sourcing from SharePoint), refreshed monthly. Clustered transactions by commodity and service type for the first commodity-level view of P-Card spend.

Lead-Time Intelligence Engine — Automation & Supplier Risk Analytics

- Architected a centralized lead-time engine in Databricks consolidating **60+** disparate vendor Excel files across **8** global commodities; the ETL layer standardized 15+ format variants, scoped with commodity and procurement teams across multiple sites.
- Automated the full monthly cycle — 6 Power Automate flows, SharePoint run-state, SharePoint-hosted Excel as system of record — saving **50+ hours/month** team-wide and cutting manual reporting effort **~90%**; co-built a Power BI executive dashboard.
- Generated 4 proprietary supplier-risk signals from a paid news and sentiment API (geopolitical disruption, financial health, OTIF volatility, MoM inflation), giving managers **6–8 weeks** of warning on supply shocks; automated crisis response with Databricks Genie, surfacing **12** high-risk supplier trends previously invisible to management.

AI & SAP Intern — Sapphire Infosystems

May 2025 – Aug 2025

- Customized the SAP FBL1N vendor line-item report over BSIK/BSAK/LFA1 tables with selection-driven views — hands-on command of relational data models and the end-to-end Procure-to-Pay cycle.
- Co-built an AI dashboard turning raw SAP data into a natural-language query layer with auto-generated visualizations (vendor-delay heatmaps, payment-trend dashboards) for non-technical business users.

PROJECTS

edgeinfer — LLM Inference Engine & Data Analyst, written from scratch

C++17 · AVX2 · Python · FastAPI · Qwen3.5 · FAISS · github.com/Arnavvs/edgeinfer

- C++ engine for Qwen3.5's hybrid Gated-DeltaNet models (AVX2 int8 kernels, recurrent-state caching), matched to Hugging Face at KL ≈ 1e-7; on a 2017 laptop CPU it beats llama.cpp and Ollama (**+11%** prefill, **+8%** decode, **25%** less RAM); lossless MTP drafting adds **+45%**.
- OpenAI-compatible server (SSE, tool calls, batching) behind a public SQL + hybrid-RAG analyst: **12/12** on a SQL-verified eval.

Wordle Distillation — Compressing a Symbolic Solver into a 0.5B LLM

PyTorch · TRL · Qwen2.5-0.5B · SFT · Constrained decoding

- Eleven controlled experiments distilling a classical Wordle solver (3.4431 avg guesses) into Qwen2.5-0.5B, which could not play the game at all untrained (91–97% failure). Reached **3.7642 guesses, 242/246 solved, 1.6% failure** on a held-out set under a feedback-consistency decoding constraint — beating both classical baselines.
- Fully reproducible: every reported number is produced by a script in the repo with seeds fixed; the two headline training runs were re-executed on different machines and produced byte-identical weights.

Noise-Aware Opinion Clustering — Dual-Agent Social Analytics Platform

MiniLM · UMAP · HDBSCAN · LangChain · Gemini · Plotly · SQLite

- Faculty-reviewed major project: 3-stage Streamlit platform clustering noisy tweets with MiniLM embeddings, UMAP reduction and HDBSCAN to **87%** cluster strength; raised average cohesion to **~94%** via per-post LLM synthetic data generation and re-embedding.
- Gemini taxonomy agent with Pydantic-constrained output builds a 3-level opinion taxonomy and scores 10 metrics per tweet, persisted to a deduplicating star-schema store; a tool-calling text-to-SQL agent answers questions and emits JSON chart specs rendered as Plotly.

Project Samarth — Natural-Language Analytics Agent over National Open Data

Python · pandas · LangChain · LangGraph · Gemini · SQLite · Streamlit

- LangGraph SQL agent that plans, writes and runs SQL over a query-logged SQLite warehouse of data.gov.in data.

EDUCATION & CERTIFICATIONS

B.Tech, Electrical & Computer Engineering — Shiv Nadar University, Greater Noida | 2022 – 2026 · IBM RAG & Agentic AI (Certified) · Coursework: AI/ML, Deep Learning, NLP, LLMs, DSA